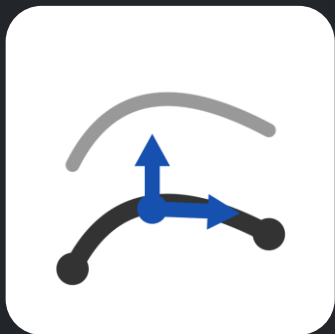




Offset edges

Offset one chain by amounts that vary along its length -- regions with ramps and dwells, blended; in-plane or out-of-plane; sharp corners handled.

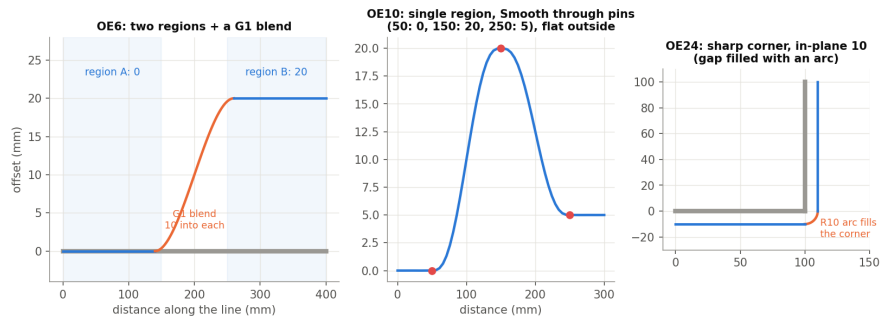


What it does

Native Offset curve is constant and in a sketch plane. Ski edges need offsets that change along the length -- a bevel that grows toward the tip, a step between two zones -- following a 3D chain.

Use it when

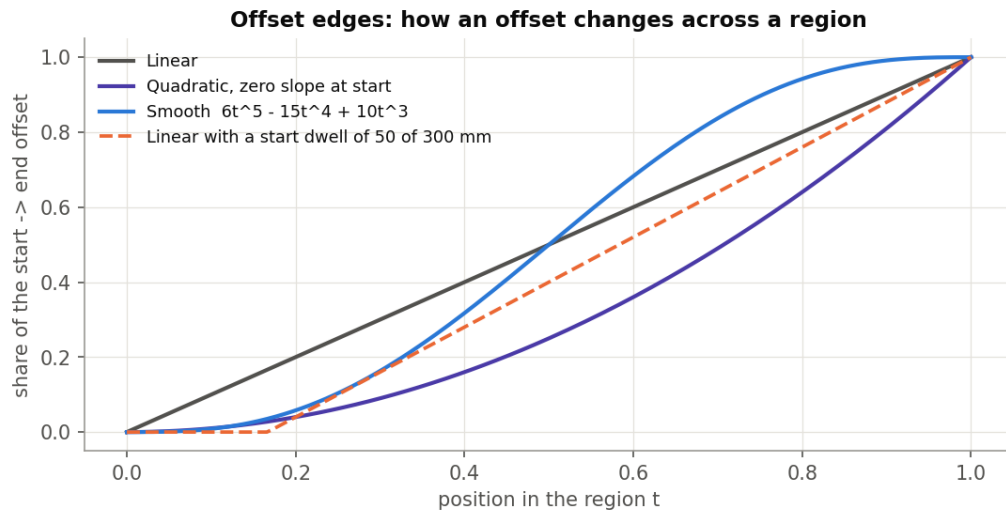
- Offsets that vary along a sidecut or edge line
- Zones with blends between them
- Out-of-plane offsets along a 3D chain



OE6 regions + blend, OE10 pins, OE24 corner.



Regions



Per region

Extent (distances from the reference or two points), start / end offset, Linear / Quadratic / Smooth, dwells.

Blends

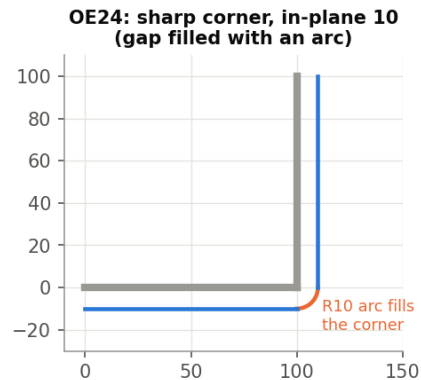
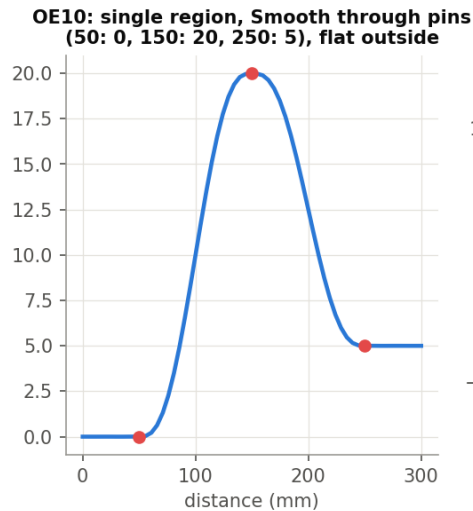
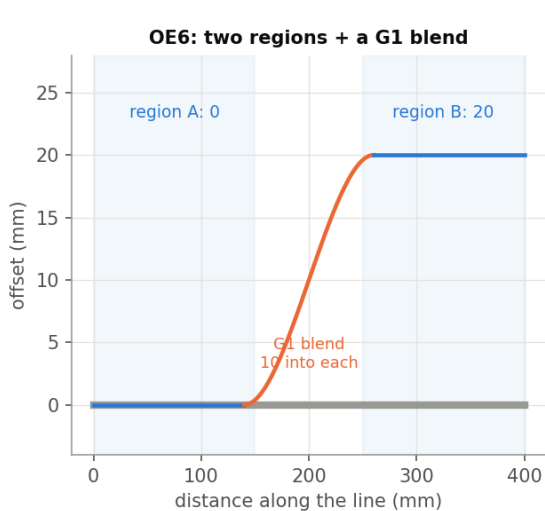
Neighbouring regions blended over a distance into each, matched to G0 / G1 / G2.

Or pins

Single region: offsets at chosen points, flat beyond the first and last.



Directions (physically)



'Normal offset'

Moves OUT of the chain's plane (a line in XY offsets along Z).

'Binormal offset'

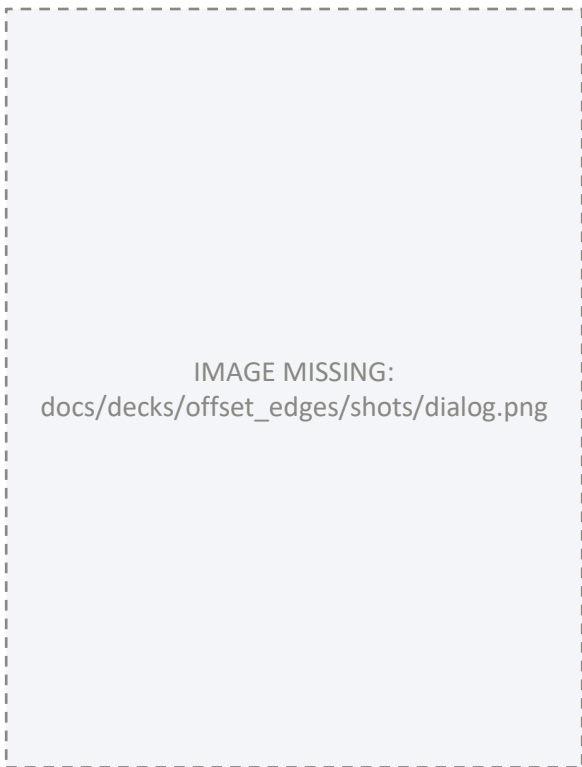
Moves WITHIN the plane, toward / away from the centre of curvature.

Names

The reverse of Frenet naming: describe results physically.



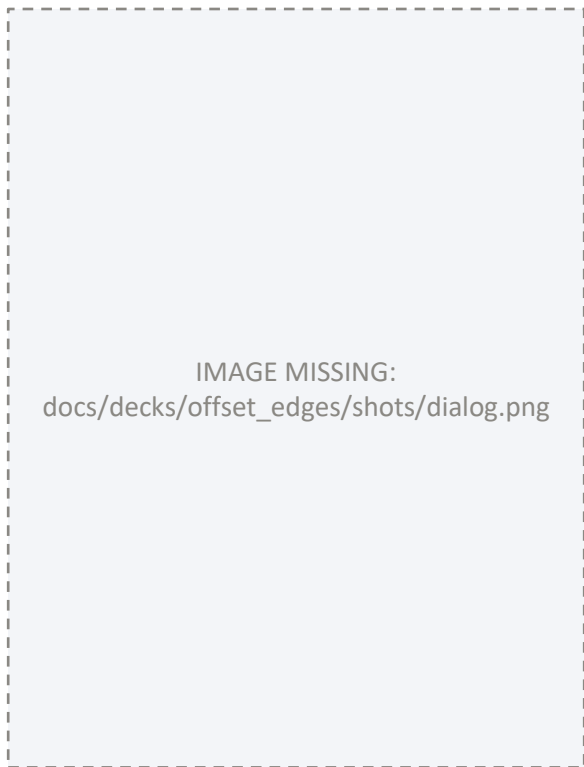
The dialog (1/2)



#	Parameter	What it means
1	Reference edges	One chain (G1, or with sharp corners).
2	Reference point	Where distance 0 is.
3	Flip direction	Reverse the distance direction (+ Show direction indicator).
4	Flip normal	Reverse the out-of-plane direction (Flip binormal likewise).
5	Source arcs	Convert to splines / Keep as arcs (concentric or arc pairs).
6	Sampling density	Points per region / blend (20).



The dialog (2/2)



#	Parameter	What it means
7	Offset layout	Multiple regions / Single region.
8	Regions	Type, name, extent, offset type, start / end offsets, dwells.
9	Intersections	Per neighbouring pair: blend, continuities, distances.
10	Approximation	Degree 3, tolerance 0.01 mm, max control points 100.
11	Debug	Show frames / regions / blends; print data.



Outputs and messages

Outputs

Key	What it holds
(wire)	One wire, split at source-edge boundaries and region / blend joints.

Messages

	Message / what to do
Info	N arc pairs, worst deviation ... Keep as arcs with varying offsets.
Warning	no regions / overlapping regions / clamped dwell or blend / region consumed by blends



Tips and limits



Keep as arcs

Constant offsets of arcs stay exact arcs (report a radius).



Gaps

Gaps between regions are gaps in the output; blending off gives a step.



Slivers

A region boundary within ~1 mm of an edge boundary can leave a sliver edge.



Where it lives

Document	example_1 (7183868d32cdc7a6ad14410d)
Tab	refSurfCreation/offsetEdges (+ engine curveMappingCore)
Tests	Offset edges tests: OE1-OE24 (all pass)
Explainer	docs/explainers/curve_mapping/curve_mapping_explained.md, section 2.5
Related	Wrap Curve, Wrap and Loft, Deform, Measure curve distance, Offset edges